THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Mathematics & Physics

Semester 2 Examinations, 2025

PHYS7045 Quantum Technologies

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•Any written or printed material is permitted; material may also be annotated.
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

Any additional written or printed material is permitted; material may also be annotated.

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x 14-Page Answer Booklet
1 x 1cm x 1cm Graph Paper

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

For Examiner Use Only

Question	Mark

Total _____

Question 1.

A code defined on nine qubits has the following stabiliser generators.

	Qubit								
Stabiliser	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
S_1	Z	Z	I	I	I	I	I	I	I
S_2	I	Z	Z	I	I	I	I	I	I
S_3	I	I	I	Z	Z	I	I	I	I
S_4	I	I	I	I	Z	Z	I	I	I
S_5	I	I	I	I	I	I	Z	Z	I
S_6	I	I	I	I	I	I	I	Z	Z
S_7	X	X	X	X	X	X	I	I	I
S_8	I	I	I	X	X	X	X	X	X

Note that in this notation, the “tensor product” symbol is suppressed, e.g $Z\,I\ldots \equiv Z \otimes I \otimes \ldots$

Part A (grade: 1-5)

1. Do you expect S_1 to commute with all the other stabilisers listed in the table? Justify your answers and show whether this holds or not.
2. A logical X_L operator for this code is a non-trivial operator that commutes with all stabilisers. Find an operator that does this and show that it commutes with the stabilisers.
3. Describe the effect of a single-qubit bit flip error (i.e. an X error) on the syndrome. Can this code detect this error? Can it correct the error? Justify your answers.
4. Describe the effect of a single-qubit phase error (i.e. an Z error) on the syndrome. Can this code detect this error? Can it correct the error? Justify your answers.
5. Explain whether the code can correct any single qubit X, Y or Z error.

Part B (Advanced: grade +0,+1,+2)

a) Find a logical Z_L operator for the code, and evaluate its commutation and anticommutation relations with the stabilisers and with X_L from part A. Do these make sense? Justify your answers.

b) What is the code distance for this code. Justify your answers.

Question 2.

Consider a pair of trapped ions, that will share collective phonon modes. The ions are prepared initially in the collective motional ground state. The following 3-pulse sequence is applied to the ions to realise an entangling gate between the two ions.

1. Red side band π-pulse on ion A that couples the ground state $|g\rangle$ and the excited electronic state $|e\rangle$.
2. A red sideband 2π-pulse applied to ion B that couples to an $|g\rangle$ to an auxiliary excited state $|a\rangle$.
3. A red sideband π-pulse applied ion A coupling the ground state $|g\rangle$ and the excited electronic state $|e\rangle$.

Part A (grade: 1-5)

1. Determine the action of step 1 of the gate to the basis states of the two-qubit system, $|gg\rangle, |ge\rangle, |eg\rangle, |ee\rangle$.
2. Starting from the result of step 1, determine the action of step 2.
3. Starting from the result of step 2, determine the action of step 3.
4. Write down the resulting 4x4 matrix for this gate and identify its action.
5. Describe how one could realise a CNOT gate using this entangling gate along with single qubit gates (you do not need to calculate the result of the circuit).

Part B (Advanced: grade +0,+1,+2)

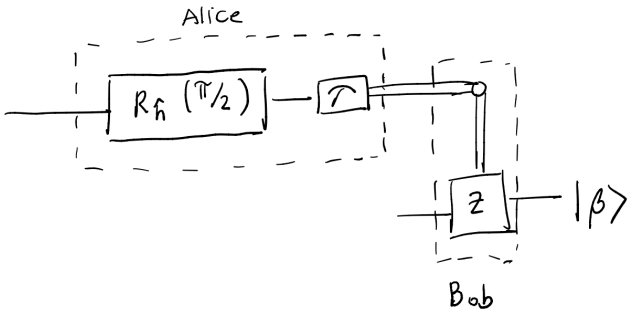
Now consider three qubits labelled A, B, C. Consider how the sequence can be modified to extend the gate to a controlled-controlled gate architecture.

a) Write down the mapping of the qubit states $|ABC\rangle_{initial} \rightarrow |ABC\rangle_{final}$ for the controlled-controlled operation.

b) Determine a modification that realises the operation by adding an additional two pulses to the sequence.

Question 3.

Consider the remote state preparation protocol represented by the following circuit:



In this protocol, Alice and Bob share in advance an entangled qubit pair in the $|\psi^-\rangle$ state, i.e. one part of the pair is with Alice, and the other part is with Bob. The goal is to remotely prepare the state $|\beta\rangle = (|0\rangle + e^{i\varphi}|1\rangle)/\sqrt{2}$ on Bob's side with Alice sending just one bit to Bob. To do this, Alice applies to her qubit a Bloch sphere rotation by $\pi/2$ about the axis $\hat{n} = (-\sin\varphi,\ \cos\varphi,\ 0)$. Then she performs a measurement in the computational basis and sends the result to Bob.

Part A (grade: 1-5)

1. Show that the state of Bob's qubit after Alice does a measurement is (up to a global phase factor) either $|\beta\rangle = (|0\rangle + e^{i\varphi}|1\rangle)/\sqrt{2}$ or $|\beta^{\perp}\rangle = (|0\rangle - e^{i\varphi}|1\rangle)/\sqrt{2}$.

2. Draw and describe an experimental setup that uses polarisation-entangled photons to achieve the remote state preparation described above.

3. The remote state preparation protocol described above is constrained in that the quantum state given by $|\beta\rangle$ lies along the equator of the Bloch sphere (i.e. $|\beta\rangle$ is an equatorial state). Consider a state $|\beta'\rangle$ which is an arbitrary point anywhere on the Bloch sphere. How would the experimental setup you drew for A.2 change to implement the remote state preparation of $|\beta'\rangle$?

4. Compare the remote state preparation to the quantum teleportation protocol. Discuss similarities and differences.

Part B (Advanced: grade +0,+1,+2)

Although quantum teleportation is often done with photons, it is also possible to do quantum teleportation using continuous variables (CV) of the electromagnetic field. Discuss advantages and disadvantages of using CV for quantum teleportation.

END OF EXAMINATION